

1

COMP

Physiology and levels of function

You should be able to: Define physiology and place a given process at the correct level of organization from molecule to organism, then state the level at which that process is best explained.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

2

COMP

Structure and function relationship

You should be able to: Predict how a change in the structure of a molecule, cell, tissue, or organ alters its function, using examples from two different organ systems.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

3

COMP

Body fluid compartments

You should be able to: State the approximate volumes of total body water, intracellular fluid, extracellular fluid, plasma, and interstitial fluid in a 70 kg adult and compare the dominant solutes of the intracellular and extracellular compartments.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

4

COMP

Compartment separation and clinical volume shifts

You should be able to: Predict the direction of water movement between compartments when extracellular osmolarity rises or falls and name a clinical situation that produces each shift.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

5

COMP

Homeostasis defined

You should be able to: Define homeostasis, regulated variable, and setpoint, and distinguish homeostasis from chemical equilibrium and from steady state.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

6

Mass balance

COMP

You should be able to: Apply the mass balance equation to a solute or to body water and calculate the intake, production, and output combination that holds the amount in the body constant.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

7

COMP

Mass flow and clearance

You should be able to: Calculate mass flow as concentration times volume flow, and clearance as rate of removal divided by plasma concentration, and explain why two patients with the same concentration can have very different delivery.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

8

COMP

Feedback loop components

You should be able to: Diagram a negative feedback loop labeling stimulus, sensor, afferent path, integrating center, efferent path, effector, and response, and trace body temperature or blood glucose through every step.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

9

COMP

Local and reflex control pathways

You should be able to: Distinguish a local control pathway from a long distance reflex pathway and classify a given response as one or the other.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

10

COMP

Negative and positive feedback

You should be able to: Distinguish negative from positive feedback by the direction of the response and give a physiological example of each, explaining why a positive feedback loop needs an outside event to end it.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

11

COMP

Feedforward control and acclimatization

You should be able to: Explain anticipatory feedforward control and acclimatization and identify which one is operating in a given scenario.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

12

COMP

Signal types and range

You should be able to: Classify a chemical signal as autocrine, paracrine, neurotransmitter, neurohormone, or hormone by its route and distance of travel.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

13

COMP

Receptor location and ligand solubility

You should be able to: Predict whether a signal molecule binds a surface receptor or an intracellular receptor from its lipid solubility and relate that to the speed and duration of the response.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

14

COMP

Signal amplification

You should be able to: Explain how a cascade amplifies a signal and estimate the size of the amplification across the steps of a given pathway.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

15

COMP

Receptor modulation

You should be able to: Define agonist, antagonist, competitive inhibition, up regulation, and down regulation, and predict the effect of chronic ligand excess on target cell sensitivity.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

16

COMP

Experimental design and controls

You should be able to: Identify the hypothesis, independent variable, dependent variable, and control condition in a physiology experiment and state what the control rules out.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

17

COMP

Measurement error and variability

You should be able to: Distinguish random from systematic error and explain why physiological measurements are repeated and averaged.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

18

COMP

Graphing and data interpretation

You should be able to: Construct a labeled graph with the independent variable on the x axis, and read slope, direction, and trend from a physiological data set.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*